higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

ELECTROTECHNICS N4

(8080074)

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09:00–12:00

Nonprogrammable calculators may be used.

This question paper consists of 6 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
ELECTROTECHNICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Questions may be answered in any order but subsections of questions must NOT be separated.
5. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 An atom is the smallest particle of a chemical element that can exist.

1.2 A current can be defined as a flow of electricity which results from the ordered directional movement of electrically charged particles.

1.3 The unit, in which the resistance is measured, is amperes.

1.4 When an increase in the temperature of material causes the resistance of the material to rise, then that material has a positive temperature coefficient of resistance.

1.5 The algebraic sum of the voltage drops in any open circuit is equal to the algebraic sum of the applied EMF's.

1.6 Right-hand grip rule states that, if a solenoid is griped in the right hand with the fingers pointing in the direction of the current flow, the outstretched thumb, if held parallel to the axis of solenoid, will point to the north end of the solenoid.

1.7 Permeability may be defined as the ease with which a magnetic field can be established in a material compared to the ease with which the same magnetic field can be established in air.

1.8 The direction of the force exerted on the conductor can be determined by applying Fleming's right-hand rule for motors.

1.9 In one of his experiments, Kirchhoff discovered electromagnetic induction.

1.10 A capacitor comprises plates and an insulating material, referred to as the dielectric.

(10 × 1) **[10]**

QUESTION 2

2.1 Nosipho Sibiya, an N4 student was given a task to connect three resistors (R_1, R_2 and R_3) in series across a supply. She then had to disconnect them and reconnect them in parallel across the same supply.

After each connection (series and parallel) she had to write down three formulas to calculate the total voltage, total current and total resistance.

2.1.1 Write down the THREE formulas from each connection. (6)

2.1.2 Write down the THREE formulas to calculate the total power. (3)

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2.2 Two batteries, A and B, which have an EMF of 200 V and 190 V and an internal resistance of 0,2 ohms and 0,5 ohms respectively, are to be charged by a generator which has an EMF of 220 V and an internal resistance of 0,1 ohms. The resistors R1 and R2 are set at 9,8 ohms and 14,8 ohms respectively.

Calculate the values of the charging currents and the current supplied by the generator. (9)

[18]

QUESTION 3

3.1 What does Faraday's law state? (3)

3.2 Three capacitors of 5, 10 and 15 µF are connected in series across a 100 volt supply.

Calculate:

3.2.1 The equivalent capacitance (2)

3.2.2 The potential difference across each capacitor (4)

3.3 What are the TWO methods of connecting field coils of DC machines? (2)

3.4 An eight-pole lap-connected armature, driven at 350 r/min, is required to generate 260 V. The useful flux per pole is 0,05 Wb. The armature has 120 slots.

Calculate a suitable number of conductors per slot. (7)

[18]

QUESTION 4

4.1 Briefly explain the following:

4.1.1 Separately-excited generator

4.1.2 Self-excited generator

4.1.3 Shunt generator

4.1.4 Series generator

4.1.5 Compound generator

(5 × 2) (10)

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4.2 A coil of 100 turns and a mean (average) area of 40 cm² per turn is rotated at 1 500 r/m in a magnetic field which has a uniform flux density of 0,05 T. The rotation is at right angles to the flux.

Calculate the following:

4.2.1 The frequency (2)

4.2.2 The period time (1)

4.2.3 The maximum value of the generated EMF (3)

4.2.4 The value of the generated EMF when the coil has rotated through 30º from the position of zero EMF (2)

[18]

QUESTION 5

5.1 Briefly explain the following AC series circuits involving RLC:

5.1.1 Consider X_L to be greater than Xc

5.1.2 Consider Xc to be greater than X_L

(2 × 3) (6)

5.2 Briefly explain the following AC parallel circuits involving RLC:

5.2.1 Consider X_L to be greater than Xc

5.2.2 Consider Xc to be greater than X_L

(2 × 3) (6)

5.3 An ideal transformer rated at 25 kVA has 500 turns on the primary winding and 40 turns on the secondary. The primary winding is connected to a 3 kV 50 Hz supply.

Calculate the following:

5.3.1 The secondary EMF

5.3.2 The primary and secondary current on full load

5.3.3 The maximum core flux

(3 × 2) (6)

[18]

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QUESTION 6

6.1 What makes it practical and economically possible for power stations to be built far from cities? (2)

6.2 Name TWO types of secondary feeder systems that are used from grid substations. (2)

6.3 What are the THREE subdivisions of Eskom? (3)

6.4 A voltmeter which has a resistance of 2 000 ohms and an ammeter which has a resistance of 0,5 ohms, were used to measure the value of two resistors. To measure the first resistor, the short-shunt method was used and the readings obtained were 10 V and 0,05 A on the voltmeter and ammeter respectively. In order to measure the second resistor, the long-shunt method was used with the readings of 5 V and 1 A obtained.

6.4.1 Calculate the percentage error in each case. (9)

6.4.2 What does the minus sign indicate in the answer to QUESTION 6.4.1? (2)

[18]

TOTAL: 100

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ELECTROTECHNICS N4

FORMULA SHEET

Any applicable formula may also be used.

1. PRINCIPLES OF ELECTRICITY

$$E = V + Ir$$

$$V = IR$$

$$R_{se} = R_1 + R_2 + \dots R_n$$

$$R_p = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots \frac{1}{R_n}}$$

$$R = \rho \frac{\ell}{a}$$

$$\frac{R_1}{R_2} = \frac{1 + \alpha_o T_1}{1 + \alpha_o T_2}$$

$$R_t = R_\theta \, [1 + \alpha_\theta \, (t - \theta)]$$

$$P = VI = I^2 R = \frac{V^2}{R}$$

$$\Phi = \frac{mmf}{S} = \frac{IN}{S}$$

$$H = \frac{IN}{\ell}$$

$$F = B\ell I$$

$$E = \frac{\Delta\Phi}{\Delta t} . N$$

$$E = B\ell v$$

$$E = \frac{L\Delta I}{\Delta t}$$

$$L = \frac{\Delta\Phi}{\Delta I} . N$$

$$Q = VC$$

$$Q_{se} = Q_t = Q_1 = Q_2 \dots = Q_n$$

$$C_{se} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \dots \frac{1}{C_n}}$$

$$Q_p = Q_1 + Q_2 + \dots Q_n$$

$$C_p = C_1 + C_2 + \dots C_n$$

2. DIRECT-CURRENT MACHINES

$$E = \frac{2Z}{c} . \frac{Np}{60} . \Phi$$

$$c = 2a$$

$$E_{gen} = V + I_a \, R_a$$

$$E_{mot} = V - I_a \, R_a$$

$$R_{start} = \frac{(V - E)}{I_a} - R_a$$

3. ALTERNATING CURRENT MACHINES

$$E_m = 2\pi BANn$$

$$e = E_m \sin \, (2\pi f . \, t \times 57{,}3)°$$

$$E_{ave} = 0{,}637 \, E_m$$

$$E_{rms} = 0{,}707 \, E_m$$

$$T = \frac{1}{f}$$

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$$f = \frac{Np}{60}$$

$$\omega = 2\pi f$$

$$Z_L = R + j\omega L$$

$$Z_c = R - j\frac{1}{\omega C}$$

$$pf = \cos\phi = \frac{R}{Z}$$

$$S = VI$$

$$P = V.I\cos\phi = I^2R$$

$$Q = V.I\sin\phi$$

4. TRANSFORMERS

$$E = 4{,}44\ f\ \Phi_m\ N$$

$$k_t = \frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

5. MEASURING INSTRUMENTS

$$R_{SH} = \frac{i_m\ R_m}{I_{sh}}$$

$$R_{se} = \frac{V}{i_m} - R_m$$

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